"I Trust It When I Understand It": Understanding How Everyday Users Negotiate Trust, Control, Safety, and Communication in Semi-Autonomous Vehicles

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Abstract

Semi-Autonomous vehicles (SAVs) are increasingly being adopted by everyday users, yet little qualitative research exists on examining how these users negotiate trust, control, and communication with these systems. Through semi-structured interviews with two Tesla drivers, we conducted an inductive thematic analysis to understand how users evaluate SAV capabilities and make sense of semi-autonomous driving features, interpret system behavior, & adjust their trust in SAV automation. Through reflexive thematic analysis, three themes emerged. First, users judge SAV competence based on comparisons to human perception capabilities. Second, users conceptualize driving as a negotiated partnership in which control of the vehicle shifts between the human and the SAV, supported by predictable override mechanisms. Third, interaction and communication modalities especially visual displays, shape cognitive load, emotional comfort, and perceived system transparency. These findings offer practical guidance for designing SAV systems that better align with user expectations, cognitive needs, and everyday driving realities.

CCS Concepts

• **Human-centered computing** → **User interface design**; **HCI theory, concepts and models**; *Interaction design theory, concepts and paradigms*; • **Social and professional topics** → *User-centered design.*

Keywords

Location Privacy, V2X Communication, Urban Traffic Simulation

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1 Introduction

Semi-Autonomous Vehicle (SAV) technology innovations have grown rapidly in the past decade and with this growth, has come a shift in public expectations around driving, safety, convenience, and human–machine collaboration [1]. While much of this improvements focus on aspects like technical performance, predictive accuracy, and societal adoption patterns, not much is understood about how everyday users make sense of SAV behavior, negotiate control, and develop confidence or hesitation when interacting with semi-autonomous systems [2]. This is a gap that needs to be resolved because as SAVs are increasingly being integrated into consumer markets, understanding how users form judgments about SAV competence, trustworthiness, control dynamics and communication modalities, becomes necessary in designing these autonomous vehicles to be safer, more usable and more trustworthy. Previous research has shown that trust in automation is influenced by many factors including system transparency, perceived reliability, communication modalities, and the user's mental model of the system's capabilities [4]. Similarly, human factors and HCI studies have emphasized that users prefer automation that augments their capabilities without making them feel incapacitated [7]. While these studies are important and relevant, research into the lived experiences of everyday SAV users especially those interacting with commercial vehicles which have been equipped with semi-autonomous features like Tesla, remain underexplored, and there is little on in-depth accounts from real SAV drivers navigating real-world contexts. Thus, there is still a lack of deeper qualitative insights into how everyday users interpret safety features, understand system boundaries, negotiate control handoffs, and evaluate communication modalities. Without these, it becomes difficult to design SAV systems that truly align with users' cognitive, emotional, and situational needs.

To address this gap, our study uses a reflexive thematic analysis approach [3] to investigate how real SAV users describe their experiences with SAV control, system communication, and how they build their trust in the SAV automation. Through semi-structured interviews with two SAV drivers who regularly use semi-autonomous features, this work provides a deeper look into how trust is calibrated, how control is shared, and how communication influences user comfort and situational awareness.

Although SAV research often emphasizes technical performance, far less is known about how everyday users make sense of the system's behaviors, especially during moments that require lightning-speed decision-making or careful interpretation of the SAV's intentions. Because SAVs are entering consumer markets, understanding these interpretations early, is necessary for safe and confident

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adoption of these vehicles. Our study was guided by the following research questions:

(1) **RQ1**: How do different forms of SAV communication (e.g., visual, auditory, haptic) shape user trust, cognitive load, and emotional comfort? We asked this question because communication is one of the few channels through which users can infer or have an idea of an SAV's internal state. Our findings show that these modalities do more than deliver information; they regulate how calm, prepared, or overwhelmed users feel. Thus, investigating the relationship between communication and trust, cognitive load and emotional comfort, helps reveal which modalities support understanding and which may unintentionally undermine trust [6, 8].
(2) **RQ2**: How do users negotiate control between themselves and the autonomous system during routine and emergency driving scenarios? Control is central to user confidence in SAVs and ultimately influences adoption of these vehicles, yet the transition between manual and automated modes is often where misunderstandings and safety failures happen. Exploring how users experience and desire control [5] especially when the SAV must intervene or when the user wants to override automation, helps clarify what kinds of shared-control models support safety and reduce anxiety.

Together, these research questions aim to explain not only what users think about SAVs but how they develop working relationships with them. These research questions evolved iteratively during analysis as themes emerged related to competence, control, and communication. Our key findings reveal that users judge SAV competence through comparisons to human limitations. We also see that they conceptualize driving as a negotiated partnership between human and machine, and evaluate communication modalities based on cognitive and emotional fit. The contribution of our work is a clearer understanding of the processes through which everyday users form trust and confidence, negotiate control boundaries, & align system communication with their expectations of safety and usability.

2 Method

2.1 Study Design and Data Collection

This study used a qualitative interview approach, specifically a reflexive thematic analysis approach, which emphasizes iterative coding, researcher subjectivity, and analytic flexibility [3] to explore how everyday users interact with and make sense of semi-autonomous vehicle systems. Two participants (AVU1 and AVU2) were recruited through convenience sampling. Both regularly operate a Tesla vehicle equipped with autonomous features such as autosteer, adaptive cruise control, and automated braking. We collected data through semi-structured interviews conducted via Zoom, each lasting approximately 30 minutes. The interview protocol included questions about perceptions of SAV communication, trust, system control, safety features, and past driving experiences. Also, the participants were encouraged to describe specific emergency scenarios where the SAV intervened as well as instances where they needed to reclaim control, or where system communication influenced their behavior. The interviews were recorded, transcribed verbatim, and then verified for accuracy, and no identifying information was retained.

2.2 Data Analysis

The analysis followed an inductive thematic approach informed by Braun and Clarke's reflexive thematic analysis framework. Our approach started with open coding, where we generated initial codes tied closely to participants' language. These codes were refined iteratively into more focused, analytic codes that captured the underlying patterns that were emerging from across the data. These codes were then compared, clustered, and interpreted to generate higher-level themes that connected user experiences across different contexts. The development of our themes emphasized analytic interpretation over surface description, particularly around how competence, control, and communication were conceptualized by the participants.

2.3 Researcher Positionality

As the researcher conducting and analyzing the interviews, my positionality shaped how I interpreted participants' experiences. My academic background in human-centered computing and my ongoing work in accessibility interaction likely positioned me to attend closely to issues of trust, transparency, and control dynamics because my experiences influence me to give more attention to how automation systems can be made safer and more trustworthy. Additionally, I also approached SAVs with a generally positive orientation toward their competence in superseeding human capabilities and their potential safety benefits, which may sensitize me to situations in which participants describe competence or predictability. Thus, I ensured to continually center participants' lived experiences rather than my own technological assumptions. Recognizing my own beliefs and expectations helped me inductively analyze the data without personal beliefs influencing my analysis. The full semi-structured interview guide is provided in the Appendix.

2.4 Limitations

While this study offers meaningful insight into early experiences with autonomous vehicle communication and control, several limitations influence the interpretation of the findings. Firstly, the sample size was very small, consisting of two participants, which limits the diversity of experiences represented. The goal of this study was depth rather than breadth, yet the views presented here cannot be generalized to all AV users or to populations with different driving habits, physical abilities, or familiarity with automation. In addition, participants were individuals who already had experience with consumer AV or advanced driver-assistance features. Their comfort with technology may differ significantly from users who are skeptical of automation or who have never interacted with AV systems. As a result, the themes identified here may overrepresent the perspectives of relatively tech-confident drivers. Also, the study relied on semi-structured interviews conducted over Zoom. While this method allowed for in-depth reflection, it did not include direct observation that often reveal behaviors, emotional reactions, and real-time breakdowns. Participants' accounts are therefore based on their remembrance of past events and may not fully capture live dynamics of collaboration with an AV system. In the same vein,

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because the interviews were not paired with real AV scenarios or simulations, participants often imagined hypothetical situations involving emergency intervention or communication breakdowns. These hypothetical reflections are valuable for getting expectations, but they may not accurately reflect how users behave under actual stress or uncertainty. Finally, the researcher's own positionality shapes the analysis. As someone trained in human-centered computing and familiar with AV design discourse, we may have interpreted participant comments through a lens that emphasizes system transparency, shared control, and communication modalities. While reflexive practices were used to mitigate this influence, the interpretive nature of qualitative analysis means that findings emerge through the interaction between participant accounts and researcher perspective. Together, these limitations do not diminish the value of the insights from this study, but rather clarify the scope of the contribution. They also point toward opportunities for future work, including larger and more diverse samples, scenario-based or simulation-based studies, and comparative analyses of users with varying levels of AV familiarity.

3 Findings

Through the qualitative analysis of interviews, we surfaced the lived experiences of the multiple stakeholders who interacted with autonomous vehicles. In this section, we present the findings of the study.

3.1 Users Judge SAV Competence Through Its Ability to Detect and Act Faster Than a Human

Participants consistently evaluated the SAV's value through its ability to perceive hazards more effectively than they could themselves. They frame the SAV's sensor and camera systems as offering expanded perception beyond their human capabilities. This expanded awareness functions as a core argument for adopting autonomous features, because the system can detect hazards the human cannot see. Rather than viewing autonomous driving as relinquishing of control, they described a sense of improved capabilities, which comes from what the vehicle can "see" and detect before the driver has the ability to do so. As AVU1 noted, driving in low-visibility nighttime environments with wildlife risks makes them actively prefer automation:

"I have not felt less in control… In fact, where we live, a lot of times I have to drive two lane roads at night and we have a lot of deer. I turn on the auto steer just because I think the car will see the deer before I do. And respond. Yeah. So, I trust the car more than I trust myself. Yeah. Because of all the cameras, my eyes can't see everything. All those cameras are seeing at the same time." — AVU1

These findings closely reflect broader research in human–automation trust [6], which shows that users tend to rely on systems when they demonstrate abilities that clearly exceed human perception or response time. In our data, this comparison between human and machine capability was grounded in specific events in which the AV detected hazards way earlier than the participant could. This suggests that trust in SAVs is primarily based on competence and rooted in observable system behavior rather than general attitudes toward technology. This comparison of human and machine-supported perception also shows how participants conceptualize the SAV's perception as an extension of their own sensory limitations. The emphasis on improved perception also appeared during high-speed or complex traffic scenarios. AVU2 recalled a moment when the AV identified a fast-approaching car in the blind spot:

"I was on the interstate, I was going to change lanes and, and the cameras on the vehicle actually picked up a car that was, you know, speeding very fast and coming up on my right hand side kind of weaving through traffic that I couldn't visually see. But the, the cameras on the right side detected the vehicle coming up and actually prevented me from turning and probably prevented a crash." — AVU2

These findings show that participants do not view AV adoption simply as a matter of convenience; rather, they describe the system as a safeguard whose perception capabilities surpass their own. This perceived "superhuman" awareness becomes the foundation for users' evaluations of whether the autonomous system is worth using at all.

Participants also described emergency intervention features as necessary conditions for adoption. They view the SAV's ability to take over during unexpected health or safety incidents as a justification for using automation. This usefulness and potential of SAVs to mitigate risks in emergencies could result in an increase in trust for SAVs. AVU2 tied the value of the AV to helping them through health vulnerabilities, noting that autonomous safety features could act in their place if a seizure occurred while driving.

"Well, on a personal note, I have a history of seizures and so one of the big selling points for me was the safety features...And so the idea being that if I were ever to have some sort of a medical emergency while driving, that the autonomous safety features in the vehicle could safely, you know, prevent any sort of a wreck or an accident and could ideally move the vehicle to the side of the road and kind of take over if I was not physically able to because of a medical emergency..." — AVU2

This shows that emergency scenarios carry emotional weight that increases the perceived value of autonomy as going beyond convenience. Participants evaluated the AV not only as a transportation tool but as a system that could protect them or others during moments of human vulnerability. In this sense, emergency handling becomes a justification for automation. When participants imagine the AV stepping in during a medical crisis, they attribute a caregiving role to the system, one they cannot reliably perform by themselves in such situations. This shows that the AV is trusted when it is imagined as stepping in to assist in cases of human limitations rather than simply performing routine driving tasks. Thus, we can say that in these cases, the AV is imagined not just as a tool but as a fail-safe safety partner capable of acting when the human cannot. Additionally, participants describe "trust" in the AV as knowledge dependent. Trust grows when the user has a clear understanding of the SAV's safety features, limitations, and how to optimize their own use of the system. In other words, trust is earned through transparency, understanding of SAV features, and informed engagement, not blind faith.

"So I think as far as building trust with the vehicle... . Understanding all the safety features and what's there, and making the best opt, kind of optimizing the vehicle and my use of the vehicle, that would be really helpful…" — AVU2

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Table 1: Codebook with Definitions and Supporting Quotes

Code	Definition	Supporting Quote
Predictive Sensing as Safety Advantage	Users describe AV sensors and cameras as extending human perception and detecting hazards earlier.	"...I trust the car more than I trust myself… my eyes can't see everything. All those cameras are seeing at the same time..." (AVU1)
Emergency Intervention as Rationale for Adoption	Users rely on AVs to mitigate risks during health or safety emergencies.	"...If I were to ever have some sort of a medical emergency… the autonomous safety features could ideally move the vehicle to the side of the road..." (AVU2)
Immediate System Takeover as Safety Requirement	Interventions must happen instantly during emergencies to prevent harm.	"...I think I would want the car to be immediate… I don't know if I would want a warning in that point..." (AVU1)
Desire for Override Mechanisms	Users want fast, simple ways to regain manual control.	"...But I think having like a one click button on the steering column that was easy to access, that would give instant control back would be very helpful..." (AVU2)
Visual Displays for Situational Awareness	Visual modality is the most trusted source of information about AV behavior.	"...I'm probably most comfortable with showing me what cars are around me on the screen..." (AVU1)
Context-Dependent Haptic/Audio Alerts	Audio and haptic cues are used selectively, mainly for urgency.	"...But usually when the car beeps... I find it annoying. It's just I'm slowing down, but it doesn't think I'm slowing down soon enough..." (AVU1)
Asynchronous Status Updates as Reassurance	Notifications outside the vehicle extend monitoring and reduce uncertainty.	"...So the majority of the time the ones that make me feel the best are the push notifications that I get, letting me know things that are happening to or around the vehicle when i'm not in it..." (AVU2)
Onboarding Needed to Build Trust	Understanding features and safety systems increases early trust.	"...So say like I bought a new vehicle... there would be a demo mode... to go through like an onboarding process visually..." (AVU2)

This all goes to show that "trust" in the SAV's competence does not arise abstractly. Instead, users form judgments about whether an SAV earns their confidence based on perceived superhuman sensing and fast reaction times as well as understanding the features of the SAV. Trust is conditional, not assumed, and emerges when the SAV demonstrates capabilities users see as surpassing human abilities, and they understand the functioning of the system. Taken together, these findings suggest that competence and transparency work together as dual conditions for early trust. Participants were willing to credit the SAV with "superhuman" sensing capabilities, but this trust was reduced when they lacked understanding of how these capabilities functioned. Trust emerged most clearly when the SAV demonstrated superior perception in real-world events, and the participant had clarity about what the system was doing and why. This indicates that trust in SAVs is neither purely just performance-driven nor is it purely based on available information; rather, it is at the intersection of perceived capability and intelligibility.

3.2 Control of the Vehicle Is Negotiated Between the Human and the Machine

Participants discussed negotiation of control and the need for dynamic modes of gaining back control after AV intervention which could increase the sense of control that they have in the SAV. In discussing system responsiveness, they emphasized that emergency intervention must be instant because delays could worsen danger. In our data, participants did not see the SAV as a replacement driver but as a partner whose authority should change depending on risk, environmental complexity, and personal readiness. This shows their expectations for how quickly and seamlessly control should shift during both routine and emergency scenarios. When the participants were asked their preference for how the SAV would take control from them if it was in manual mode and they were in an emergency scenario, their preference was:

"Yeah, I think I'd want a warning or a countdown. Well, I mean, even in that situation, a few seconds could make a difference. Yeah, let me think about that question… I needed time to think because I changed my answer. Okay, so if the car is sensing there's an emergency, there's probably already been a few seconds and a few seconds more could be really bad. So, I think, you know, if my head drops and it's... I'm not responding, I think I would want the car to be immediate. I don't know if I would want a warning in that point because like I said, there would already have been some time that the car would have sensed a problem. So, it probably needs to respond right away without a warning." — AVU1

"I would want it to be instant… That's correct. And maybe there would be even multiple ways to gain control back. But I think having like a one click button on the steering column that was easy to access, that would give instant control back would be very helpful." — AVU2

This preference for immediacy highlights the psychological dynamics of relinquishing control. Participants were willing to allow

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automation to intervene only if the system demonstrated responsiveness that matched or exceeded their own reflexes. While participants valued the SAV's ability to intervene during safety-critical events, they also expressed a strong desire to retain the ability to reassert control at any moment. So, while they wanted the vehicle to be able to take control and switch from manual mode to auto mode, in the event that there was an emrgency scenario, they also wanted to be able to gain control of the SAv and switch from auto mode to manual mode. This functionality being available, especially in moments of need like the need to maneuver out of heavy traffic, could increase the user's sense of control by being able to commandeer the AV anytime they want.

Yes. If I'm in very, very heavy traffic, stop and go traffic. If I'm on 85 and the traffic is, you know, really thick... I would rather just drive myself than use autosteer." — AVU1

While they expected instant machine action in emergency scenarios without a warning or confirmation prompt, yet in everyday driving, they wanted equally fast ways to override SAV automation. AVU2 described the ideal mechanism:

"… But I think having like a one click button on the steering column that was easy to access, that would give instant control back would be very helpful... But I think having access to quickly override any sort of autonomous control via the steering column without confirmation. I would, I would like…" – AVU2

This need for a rapid override mechanism supports a conceptualization of driving as a shared control model in which authority moves fluidly between human and the machine. This balance of control is also influenced by context. AVU1 noted that when there are no incidents or need for intervention, they prefer to have control over putting the SAV in auto or manual mode, but they relinquish control strategically when environmental conditions reduce their driving confidence.

"But on a regular day when I'm healthy and everything's fine, I still want to be in control of turning on auto steer on and off. Does that make sense?" – AVU1

Thus, control becomes a negotiated and situation-dependent partnership rather than a fixed mode of operation.

3.3 Communication Modality Shapes Cognitive Load, Emotional Comfort, and Perceived System Transparency

Participants evaluated communication modalities not only as interfaces but as mechanisms shaping emotional comfort and cognitive load. Visual displays were preferred over other modalities like voice because they provide transparency and a real-time depiction of what the AV "sees," which reduces uncertainty supports user confidence. Participants rely on visual displays to maintain cognitive and spatial orientation during driving. AVU1 described visual displays as "my security," explaining that seeing the environment rendered on screen confirmed that the AV was perceiving hazards correctly.

"I'm probably most comfortable with showing me what cars are around me on the screen...So, I would say the camera is probably the most security because they're showing me what's on the screen. So it'd be that combination of the screen display from the cameras." — AVU1

Audio and haptic notifications were less preferred in comparison to visual displays:.

"Voice commands or voice confirmations are fairly low on the priority list for me. I would prefer visual notifications like a pop up or haptic or an actual kind of display pop up or message to read as opposed to voice commands or voice notifications." — AVU2

Audio and haptic notifications were valued for urgent situations but described as irritating or burdensome when used frequently. Participants wanted the communication modality to match the situational context. Thus beeps and haptics were preferred for emergencies, and visual cues for routine navigation. This fine-grained distinction suggests that communication is not merely informational but deeply tied to how users maintain situational awareness and regulate attention.

"But if... maybe my mind's wandering a little bit and there's somebody stopping really quick in front of me. The sound of the beep is also...Yeah, I need that at that point in time, if, you know, I need to slow down fast... But usually when the car beeps, I'm aware, and so I find it annoying. It's just I'm slowing down, but it doesn't think I'm slowing down soon enough. Um, so I like the visual the best." — AVU1

Communication outside the vehicle was also discussed. AVU2 emphasized that asynchronous notifications provided reassurance by extending awareness beyond the driving context. Receiving push notifications about system status and environmental events made them "feel the best," reinforcing perceptions of the AV as a continuously monitoring partner.

"So far? It has been actually the cell phone push notifications. So I don't drive the vehicle that often,... So the majority of the time the ones that make me feel the best are the push notifications that I get, letting me know things that are happening to or around the vehicle when I'm not in it." — AVU2

To also reduce anxiety and uncertainty when operating the AV for the first time, participants discussed the need for a structured, guided explanations of AV features.

"I almost wonder if there would be, you know, I like it like a demo of some type. So say, like, I bought a new vehicle, you know, either new or used, but it was new to me, and there would be a demo mode to where I could sit in the vehicle, connect my app, and the visual display would go through like an onboarding process visually through the display to where it walked me through the setup of. Here's every single safety feature that you can turn on, turn off. Here's the preferences that you can choose from. Here is what this could do. Here's what it might, you know, here's what it currently looks like, and then walk you through the entire setup and kind of, kind of a white glove, onboarding, concierge, onboarding, whatever you want to call it, so to speak, as opposed to having to figure it out on your own, if that makes sense...." — AVU2

4 Discussion

The findings from this study show that early experiences with autonomous vehicles are influenced by three interdependent processes: evaluating system competence, negotiating shared control, and interpreting communication modalities. Together, these processes show how users build trust, develop confidence, and construct a mental model of the AV as a collaborative partner rather

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than a replacement driver. In this section, we interpret these findings, relate them to broader research in human–automation interaction, and outline the implications for AV design, policy, and future deployment.

4.1 Perceived System Competence and the Conditions of Trust

The way users assess SAV competence aligns with prior research showing that trust in automation grows when systems reliably perform beyond human limitations, especially in detection and response time. Participants relied on the SAV's enhanced sensing capabilities to identify risks they themselves might miss. This perception of "superhuman awareness" directly shaped their trust and willingness to accept autonomous features. However, the findings are nuanced and don't just convey simple narratives of trust. Participants did not trust the SAV because it had advanced capabilities; rather, trust was conditional and they still required system transparency. Also, without structured onboarding, participants expressed uncertainty about what the SAV could and could not do. This insight contributes to the literature by highlighting trust as a knowledge-dependent construct. Participants wanted explanations, demonstrations, and clear descriptions of safety features before fully embracing autonomous functionality. This suggests that manufacturers and policymakers should treat onboarding as a safety component rather than an optional convenience. This is necessary as users might not be motivated to use SAVs if they lack clear, guided exposure to system capabilities. Structured onboarding, demonstrations, and transparent feature explanations could improve both safety and consumer confidence.

4.2 Shared Control as a Preferred Model of Human–Automation Interaction

Our findings on control preferences are supported by broader research on shared autonomy, where users prefer a partnership rather than full automation. Participants consistently described wanting flexible control boundaries that shift according to context such as dense traffic, emotional stress, perceived risk, or unexpected events. This introduces a nuanced contribution: users do not simply want "control" or "automation," but the ability to be able to dynamically negotiate between both of them. Immediate override mechanisms were seen as important, not optional. This suggests that SAVs designed around non-instant handover protocols may unintentionally impact user confidence, especially in safety-critical moments. The findings also raise important considerations for policymakers. If users expect immediate manual override, regulatory frameworks must address how SAVs will balance their safety logic with human authority. For example, should a human be allowed to override an automated emergency maneuver if they believe they can act faster? These questions have implications for liability, training requirements, and consumer expectations. Thus, future AV systems must support dynamic, shared control that is sensitive to situations, rather than the binary modes of manual vs. autonomous operation.

4.3 Communication Modalities as Emotional and Cognitive Regulators

Communication modality emerged as a main factor in shaping emotional comfort, situational awareness, and cognitive load. Participants preferred visual feedback because it allowed them to build a mental model of the SAV's reasoning. Audio and haptic cues were useful only when conveying urgency and overuse of these cues caused irritation or anxiety. This finding expands prior work by showing that multimodal communication is not simply an interface preference but an emotional and psychological regulator. The SAV's communication style influences if users interpret system behavior as predictable, competent, and aligned with their needs or not. Poorly calibrated communication risks undermining trust even if the underlying automation is functioning correctly. These insights have direct implications for interaction design. AVs should adapt communication modality to situational context like reserving audio alerts for urgent events but relying on visual displays for routine information. They should also account for user state (stress, distraction level, experience) to avoid unnecessary cognitive burden.

4.4 Broader Implications for Design, Training, Regulation, and Deployment

Taken together, the findings point toward a model of SAV adoption grounded not in blind trust but in supported confidence. Several implications emerge; **For Designers**: SAVs must explain their "reasoning" through visual, comprehensible displays. Onboarding guides should also be designed to walk users through safety features, limitations, and best practices. Also, override mechanisms must be physically accessible, intuitive, and responsive. **For Older Adults and New Users**: the provision of structured onboarding reduces anxiety and promotes safer early use. Transparent communication also helps to build stable expectations of SAV behavior. **For Industry**: SAV manufacturers may benefit from standardized onboarding protocols similar to aviation checklists or medical device safety training. **For Policy Regulators**: Regulators could require documentation or demonstration of safety features as part of AV delivery. Liability frameworks may also be encouraged to account for dynamic shared control, not simple manual vs. automated modes. Overall, the findings reinforce that SAVs must be designed not simply as technical systems but as collaborative partners that communicate clearly, act predictably, and support the human in maintaining situational and emotional control. As SAVs continue to enter everyday mobility, these insights can inform more accessible, trustworthy, and user-centered deployment strategies.

A Interview Protocol

A.1 Study Overview

This semi-structured interview study explores how users experience communication and collaboration with autonomous vehicles (AVs). The goal is to understand how different communication modalities (visual, auditory, and haptic) shape trust, cognitive load, and sense of control during both routine and emergency scenarios.

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A.2 Eligibility Criteria

- Adults aged 18 or older.
- Prior experience with any form of vehicle automation (e.g., ADAS features, Tesla Autopilot).

A.3 Data Collection Procedure

Interviews were conducted over Zoom and lasted approximately 30–45 minutes. Participants were informed of confidentiality and voluntary participation. No identifying information was collected. Questions were asked in a flexible order, and probing questions were used when appropriate to elicit richer detail.

A.4 Interview Guide

1. Experience and Context

- Can you tell me about your experience driving or riding in an autonomous vehicle?
- Probe: What excites or concerns you most about AVs?

2. Information Exchange

- When you interact with the AV, what kinds of information does it communicate to you, and how does it do so?
- Probe: Are these messages visual, spoken, or tactile?

3. Preferred Communication Modes

- What forms of communication (voice, screens, sounds, vibrations) make you feel most comfortable or confident?

4. Trust in Communication

- Can you describe a situation where the AV's communication made you feel trusting or distrusting of its decisions?
- Probe: What influenced your level of trust?

5. Sense of Control

- Have you ever felt less in control while using the AV? What contributed to that feeling?

6. Control Transition Preferences

- In what types of driving situations would you want to take control from the AV?
- Probe: How should the AV notify you that control is being transferred (sound, visual cue, vibration)?

7. Emergency Scenarios

- In an emergency, would you prefer the AV to remain in control or for you to take over? Why?
- Follow-up: What type of alert or signal would help you respond effectively?

8. Timing of Handover

- Would you want the control transfer to happen instantly or with a countdown? How much warning feels comfortable?

9. Accessibility and Personal Preferences

- Do certain communication modes (voice, visuals, haptics) feel more or less accessible or comfortable for you? Why?

10. Design Recommendations

- If you could give AV designers one piece of advice for improving communication and collaboration, what would it be?

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